

Instant Coffee Tray Service SOP (1x Cup)

Prepare one cup of instant coffee per episode. Work in this order: move the saucer, place the cup, add coffee, pour water, stir, return the spoon, then move the cup and saucer simultaneously to the target tray.

Arm assignments are fixed:

- **Left gripper:** moves the cup to center, holds the sachet for handoff, lifts the spoon at its middle for handoff, stabilizes the saucer during stirring, and returns the spoon to the staging tray.
- **Right gripper:** moves the saucer to center, takes, empties, and returns the sachet, handles the kettle, and takes the spoon at its handle end for stirring.
- **Final transfer:** both arms move the service simultaneously. The left gripper holds the saucer and the right gripper holds the cup handle.

Always grasp the cup by its handle. Never touch hot water or the kettle body, lid, or nozzle.

Setup

Start only after all input objects are separated on the left staging tray and both checklists pass.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are recording. The staging tray, center, kettle, and target tray are visible.
2. The kettle nozzle and cup rim will remain visible during the pour.
3. Both arms are home with grippers open.
4. The kettle is closed and seated on its stand, handle toward the right arm.
5. The kettle contains at least 500 mL of water at 90–95 °C.
6. The table is clean, dry, and clear except for task materials.

Materials checklist

1. One clean, dry staging tray is on the left.
2. These objects are separated on the staging tray with visible gaps and no overlap:
 - one empty, dry 200 mL cup, upright;
 - one dry saucer, face up;
 - one open coffee sachet, flat with its opening facing right; and

- one dry teaspoon, bowl down.
- 3. One empty, dry target tray is in the right output zone.
- 4. The cup fits the saucer, and the cup and saucer fit fully on the target tray.

Workspace layout

- **Left:** staging tray with the separated cup, saucer, sachet, and spoon
- **Center:** assembly and coffee-preparation area
- **Back center:** kettle and stand
- **Right:** target tray

Vocabulary

- **Cup-handle grasp:** grasp the handle; never grasp the cup rim or body.
- **Centered cup:** an even ring of saucer is visible around the cup base.
- **Pour position:** the kettle nozzle is above the cup with a visible gap from the rim.
- **Dissolved coffee:** no granules or clumps are visible.
- **Unsafe spill:** hot water leaves the cup.

Steps

Step 1: Move the saucer, then the cup

Goal: the cup is centered on a stable saucer in the center.

1.1 Move the saucer

- With the **right gripper**, grasp the near saucer rim, lift it from the staging tray, and place it flat in the center.
- Confirm the saucer is stable. Reseat it once if it rocks.

1.2 Place the cup

- With the **left gripper**, grasp the cup handle.
- Lift the cup from the staging tray and lower it into the saucer well.

Check: the saucer is stable and the cup is centered. Regrasp the cup handle and reseat the cup once if needed. Do not slide either object.

Step 2: Hand off, empty, and return the sachet

Goal: the coffee is in the cup and the empty sachet is back on the staging tray.

2.1 Hold and hand off the sachet

- With the **left gripper**, hold the sachet at its lower end on the staging tray, opening up.

- With the **right gripper**, grasp the sachet body below the opening.
- Release the left gripper only after the right gripper has a secure grasp.

2.2 Empty and return the sachet

- With the right gripper, lift the sachet from the staging tray, keeping the opening up.
- Center the sachet opening inside the cup rim and tilt the sachet until the coffee pours out.
- Hold the sachet tilted until the flow stops, then hold it 2 more seconds before righting it.
- If granules remain in the sachet, tilt and pour once more. Never pour a third time.
- With the right gripper, return the empty sachet flat to its original staging-tray position, opening facing right.

Check: the sachet is empty and returned. Record any spill. Do not sweep granules into the cup.

Step 3: Pour hot water

Goal: fill the cup and return the kettle.

3.1 Lift and position

- With the **right gripper**, grasp the kettle handle and lift it straight off the stand.
- Carry the kettle upright until the nozzle is in the pour position.

3.2 Pour

- Tilt and pour into the center of the cup until the cup is filled, stopping below the rim.
- Return the kettle upright over the cup and let it drain before moving it away.

3.3 Return the kettle

- Carry the kettle upright to its stand, seat it fully, then release it.

Check: the kettle is stable and the cup is filled. Reseat a rocking kettle once.

Spill branch: if hot water leaves the cup, return the kettle upright and seat it if the path is safe. Clear both arms and end the episode. Do not move the cup or wipe hot water while recording. Use the station emergency procedure if the return path is unsafe.

Step 4: Stir and return the spoon

Goal: dissolve the coffee and return the spoon to the staging tray.

4.1 Lift and hand off the spoon

- With the **left gripper**, grasp the middle of the spoon and lift it from the staging tray.
- Present it at center. With the **right gripper**, grasp the end of the spoon handle.
- Release the left gripper only after the right gripper has a secure grasp.

4.2 Stir and inspect

- With the **left gripper**, hold the far saucer rim.
- With the **right gripper**, lower the spoon bowl into the coffee.
- Stir five full circles clockwise without touching the cup, then lift and drain the spoon over the cup for 2 seconds.
- If granules remain, repeat five clockwise circles once. Never perform a third cycle.

4.3 Hand back and return the spoon

- Release the saucer with the **left gripper**, then grasp the middle of the spoon.
- Release the right gripper only after the left gripper has a secure grasp.
- With the left gripper, return the spoon bowl down to its original staging-tray position.

Check: the coffee is dissolved, the cup remains centered, and the spoon is back on the staging tray.

Step 5: Move the cup and saucer simultaneously

Goal: the cup and saucer are stable on the target tray.

5.1 Grasp and lift

- Close both grippers simultaneously: the **left gripper** on the near saucer rim and the **right gripper** on the cup handle.
- Lift the cup and saucer simultaneously. Keep the saucer level and cup upright.

5.2 Transfer and release

- Move both arms simultaneously, carrying the objects to the target tray on separate, non-crossing paths.
- Lower the saucer into the marked position on the target tray while the right gripper continues holding the cup.
- When the saucer is stable, release the saucer, then release the cup.

Check: the cup is centered, the saucer is stable, and both are fully inside the target tray. Reseat the service once if it overhangs.

Step 6: End the episode

1. Confirm the cup and saucer are stable on the target tray.
2. Confirm the kettle is seated and the sachet and spoon are back on the staging tray.
3. Return both arms home with grippers open, then end data collection.

After the episode: reset the workspace

Reset with recording off.

1. Let the drink and kettle reach a safe handling temperature.
2. Remove the service and discard the empty sachet.
3. Wash and dry all task objects, the table, and the kettle exterior.
4. Refill and heat the kettle, restage the four separated inputs, and run Setup.

SOP violations

These are actions that break the SOP and are reviewed in the side-by-side review tool.

How to record a violation

For each violation, record:

- the **start timestamp**;
- the **violation name** exactly as written; and
- the **SOP rule broken**.

Use the most specific label. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag the violation and retain the episode unless project policy excludes it. Record system and material failures separately.

Violations

Violation: Required sequence not followed

- **Visible cue:** the cup moves before the saucer is stable in the center; the cup and saucer are moved from the staging tray at the same time; coffee, water, or stirring happens out of order; or the service moves before the spoon is returned.
- **SOP rule broken:** Steps 1–6 (saucer, cup, sachet, kettle, spoon, final transfer, then end).
- **Coaching note:** follow the fixed sequence from saucer placement through final transfer.

Violation: Prescribed arm, gripper, or tool assignment not followed

- **Visible cue:** the left gripper moves the saucer, lifts or pours the sachet, handles the kettle, or stirs; the right gripper moves the cup; or the final transfer uses the wrong grips.
- **SOP rule broken:** Steps 1 to 5, left holds the sachet for handoff, right lifts, pours, and returns it, and all other actions use their written grippers.
- **Coaching note:** follow the written gripper assignment for each object and handoff.

Violation: Object handled or transported incorrectly

- **Visible cue:** an object is dragged, tilted when it should stay level, moved on a crossing path, or released before it is stable.
- **SOP rule broken:** Steps 1–5 (lift before moving, follow the stated path and orientation, and release only when stable).
- **Coaching note:** use the stated grasp, path, orientation, and release condition.

Violation: Object dropped or released uncontrollably

- **Visible cue:** an object falls or is released outside its destination.
- **SOP rule broken:** Steps 1–5 (maintain control and release only at the assigned destination).
- **Coaching note:** confirm the grasp and landing area before moving.

Violation: Required check, retry, or recovery not followed

- **Visible cue:** a failed check is ignored, a retry limit is exceeded, or an unwritten recovery is used.
- **SOP rule broken:** Steps 1 to 5, perform each check and use only its allowed retry or recovery.
- **Coaching note:** stop at each check and use only its written recovery.

Violation: Cup not grasped by the handle

- **Visible cue:** a gripper closes on the cup rim or body instead of its handle.
- **SOP rule broken:** Steps 1.2 and 5.1, always grasp the cup handle.
- **Coaching note:** always grasp the cup handle.

Violation: Cup or saucer placed incorrectly

- **Visible cue:** the saucer rocks, the cup is off-center, or either object is slid instead of lifted for correction.
- **SOP rule broken:** Steps 1.1–1.2 (place the saucer flat and the cup centered; lift to correct).
- **Coaching note:** place the saucer flat and lift the cup by its handle to recenter it.

Violation: Sachet handled incorrectly

- **Visible cue:** the opening points down before reaching the cup; the sachet is shaken or flicked instead of poured; granules empty outside the cup; coffee remains inside; a third pour is made; or the sachet is not returned flat.
- **SOP rule broken:** Step 2.2 (use the right gripper to pour the sachet empty into the cup, with one extension only when needed, and return the sachet to its original staging-tray position).
- **Coaching note:** keep the opening up during transit, pour over the cup, then return the sachet.

Violation: Spill, debris, or waste handled incorrectly

- **Visible cue:** granules land outside the cup; granules are swept into the cup; or hot water is wiped, or the hot service is moved, after a spill.
- **SOP rule broken:** Step 2.2, Step 3 spill branch, and Step 4.2 (keep material in the cup and follow the written spill response).
- **Coaching note:** do not return spills; follow the hot-water spill branch.

Violation: Pour safety condition not followed

- **Visible cue:** the kettle tilts before reaching the cup, the nozzle touches the cup, or the kettle is released before seating.
- **SOP rule broken:** Steps 3.1–3.3 (carry upright, keep a nozzle gap, and seat the kettle before release).
- **Coaching note:** carry upright, keep a gap, and seat before release.

Violation: Hot water spilled

- **Visible cue:** water leaves the cup at any point — during the pour, while stirring, or during the final transfer — and lands on the saucer, the table, the target tray, or another object.
- **SOP rule broken:** Step 3.2 and the Step 3 spill branch (all water stays in the cup; an unsafe spill ends the episode).
- **Coaching note:** pour slowly into the center of the cup and stop below the rim; if water leaves the cup, seat the kettle, clear both arms, and end the episode.

Violation: Handoff not completed correctly

- **Visible cue:** the left gripper does not hold the sachet before the right grasps it; the right lifts the sachet before the left releases; the spoon grips are incorrect; or a sending gripper releases before the receiving grasp is secure.
- **SOP rule broken:** Steps 2.1, 4.1, and 4.3 (complete each handoff with both grippers holding before the sending gripper releases).
- **Coaching note:** establish the receiving grasp before releasing the spoon.

Violation: Stirring performed incorrectly

- **Visible cue:** the left gripper stirs; fewer than five circles are made; any circle runs counterclockwise; the spoon touches the cup; the left gripper does not hold the saucer; or a third cycle is performed.
- **SOP rule broken:** Step 4.2 (left stabilizes the saucer; right stirs five clockwise circles, with one extension only when needed).
- **Coaching note:** hold with the left, stir clockwise with the right, and extend only once if needed.

Violation: Spoon not returned correctly

- **Visible cue:** the spoon remains in the cup or center, or is not bowl down in its original staging-tray position.
- **SOP rule broken:** Step 4.3 (return the spoon bowl down to its original staging-tray position).
- **Coaching note:** drain the spoon, then return it to its original staging-tray position.

Violation: Final transfer not performed with both arms

- **Visible cue:** the cup and saucer move separately; one arm lifts or carries before the other; one arm performs the transfer; the left gripper does not hold the saucer; or the right gripper does not hold the cup handle.
- **SOP rule broken:** Steps 5.1–5.2 (grasp, lift, and carry simultaneously, with left holding the saucer and right holding the cup handle throughout the final transfer).
- **Coaching note:** grasp both objects and move them simultaneously with their assigned arms.

Violation: Final transfer or target-tray placement incorrect

- **Visible cue:** the arm paths cross; the service tilts; the cup or saucer is released early; or either object remains outside the target tray.
- **SOP rule broken:** Steps 5.1–5.2 (use separate paths, keep both objects controlled, and release only after the saucer is stable on the target tray).
- **Coaching note:** keep both objects controlled until the saucer is stable on the target tray.

Violation: Workspace or tool state not restored

- **Visible cue:** the kettle is off its stand or the sachet or spoon is not in its staging-tray position.
- **SOP rule broken:** Steps 2.2, 3.3, 4.3, and 6 (restore the sachet, kettle, and spoon before homing).
- **Coaching note:** restore all required final states before homing.

Violation: Incorrect episode ending

- **Visible cue:** recording ends before final checks, with arms away from home, or with closed grippers.
- **SOP rule broken:** Step 6 (complete final checks, home both arms with open grippers, then end).
- **Coaching note:** complete final checks, home both arms, then end.

Non-violation failures: log and discard an unusable episode for a recording, camera, robot, kettle, or fixture failure; defective material; or an unexpected safety hazard.

Annotation subtasks (from SOP)

1. Move the saucer to the center with the right gripper
2. Place the cup on the saucer with the left gripper
3. Hand the sachet from the left gripper to the right gripper
4. Empty and return the sachet with the right gripper
5. Lift, pour, and return the kettle with the right gripper
6. Hand the spoon from the left gripper to the right gripper
7. Stir and inspect the coffee
8. Hand the spoon back and return it to the staging tray
9. Grasp and move the cup and saucer simultaneously
10. Place the service on the target tray
11. Return both arms home and end the episode